

Module 7: Exercise Multinomial Dirichlet Conjugacy

Rebecca C. Steorts

Agenda

- ▶ Dirichlet distribution
- ▶ The Dirichlet-Multinomial

Dirichlet

A Dirichlet distribution¹ is a distribution of the $K - 1$ -dimensional probability simplex²

$$\Delta_{K-1} = \{(\pi_1, \dots, \pi_k) : \pi_k \geq 0, \sum_k \pi_k = 1\}.$$

We say that $(\pi_1, \dots, \pi_k)$ is Dirichlet distributed:

$$(\pi_1, \dots, \pi_k) \sim \text{Dir}(\alpha_1, \dots, \alpha_k)$$

if

$$p(\pi_1, \dots, \pi_k) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_{k=1}^K \pi_k^{\alpha_k-1}.$$

¹This is the multivariate version of the Beta distribution.

²In geometry, a simplex is a generalization of the notion of a triangle or tetrahedron to arbitrary dimensions.

Dirichlet distribution

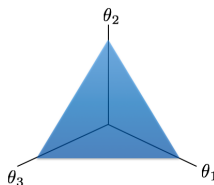
Let

$$\theta \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$$

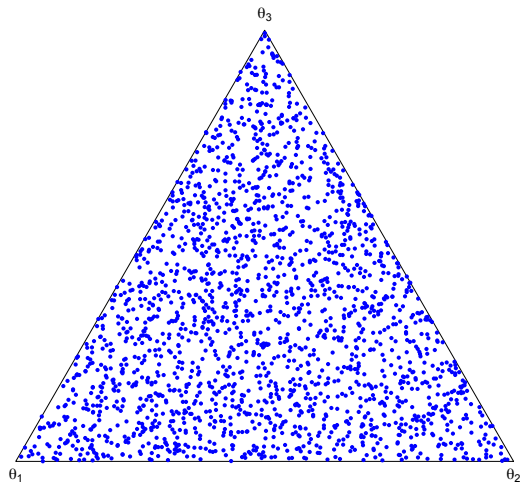
where the probability density function is

$$p(\theta \mid \alpha) \propto \prod_{k=1}^m \theta_k^{\alpha_k - 1},$$

where $\sum_k \theta_k = 1, \theta_i \geq 0$ for all i



Dirichlet distribution



Dirichlet distribution

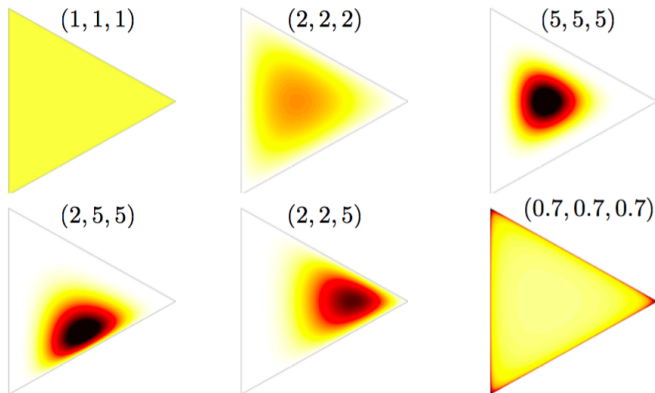


Figure 1: Far left: We get a uniform prior on the simplex. Moving to the right we get things unimodal. On the bottom, we get distributions that are multimodal at the corners.

Multinomial-Dirichlet

In order to proceed with the lab, we'll need to learn about the Multinomial or Categorical distribution.³

³This is the multivariate generalization of the Binomial distribution.

Multinomial or Categorical distribution

Assume $X = (x_1, x_2, \dots, x_n)$ where $x_i \in \{1, \dots, m\}$.

1. n : number of observations.
2. m : number of categories or bins.

Assume $\theta = (\theta_1, \dots, \theta_m) \geq 0$, where $\sum_i \theta_i = 1$.

Assume that

$$X \mid \theta \stackrel{ind}{\sim} \text{Multinomial}(\theta)$$

or

$$X \mid \theta \stackrel{ind}{\sim} \text{Categorical}(\theta)$$

$$P(X_i = j \mid \theta) = \theta_j$$

where $j \in \{1, \dots, m\}$ and $i = 1, \dots, n$.

Binomial Special Case

Consider $j \in 1, 2$. Let $i = 1, \dots, n$.

Then $\theta = (\theta_1, \theta_2) \geq 0$, where $\theta_1 + \theta_2 = 1$.

$$X \mid \theta \stackrel{ind}{\sim} \text{Categorical}(\theta).$$

1. Suppose that $j = 1$. Then $P(X_i = 1 \mid \theta) = \theta_1$.
2. Suppose that $j = 2$. Then $P(X_i = 2 \mid \theta) = \theta_2$.

This is equivalent to $X \sim \text{Binomial}(n, \theta_1)$.

Multinomial or Categorical distribution

Consider $j \in 1, 2, 3$. Let $i = 1, \dots, n$.

Then $\theta = (\theta_1, \theta_2, \theta_3) \geq 0$, where $\theta_1 + \theta_2 + \theta_3 = 1$.

$$X \mid \theta \stackrel{ind}{\sim} \text{Categorical}(\theta).$$

1. Suppose that $j = 1$. Then $P(X_i = 1 \mid \theta) = \theta_1$.
2. Suppose that $j = 2$. Then $P(X_i = 2 \mid \theta) = \theta_2$.
3. Suppose that $j = 3$. Then $P(X_i = 3 \mid \theta) = \theta_3$.

Example: Extend heights example to three categories of gender, which might be male, female, and other.

Conjugate prior (Dirichlet)

$$\boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\alpha})$$

Recall the density of the Dirichlet is the following:

$$p(\boldsymbol{\theta} \mid \boldsymbol{\alpha}) \propto \prod_{j=1}^m \theta_j^{\alpha_j-1},$$

where $\sum_j \theta_j = 1, \theta_i \geq 0$ for all i

Likelihood

Define the data as $\mathbf{X} = (x_1, \dots, x_n)$, $x_i \in \{1, \dots, m\}$. Consider

$$p(\mathbf{X} \mid \theta) = \prod_{i=1}^n P(X_i = x_i \mid \theta) \quad (1)$$

$$= \prod_{i=1}^n \theta_{x_i} = \theta_{x_1} \times \theta_{x_2} \times \theta_{x_n} \quad (2)$$

$$= \prod_{i=1}^n \prod_{j=1}^m \theta_j^{I(x_i=j)} = \prod_{j=1}^m \prod_{i=1}^n \theta_j^{I(x_i=j)} \quad (3)$$

$$= \prod_{j=1}^m \theta_j^{\sum_i I(x_i=j)} \quad (4)$$

$$= \prod_{j=1}^m \theta_j^{c_j} \quad (5)$$

where $c = (c_1, \dots, c_m)$, $c_j = \#\{i : x_i = j\}$.

Likelihood, Prior, and Posterior

$$p(\mathbf{X} \mid \boldsymbol{\theta}) = \prod_{j=1}^m \theta_j^{c_j}$$

$$P(\boldsymbol{\theta}) \propto \prod_{j=1}^m \theta_j^{\alpha_j - 1} I(\sum_j \theta_j = 1, \theta_i \geq 0 \forall i)$$

Then

$$P(\boldsymbol{\theta} \mid \mathbf{X}) \propto \prod_{j=1}^m \theta_j^{c_j} \times \prod_{j=1}^m \theta_j^{\alpha_j - 1} I(\sum_j \theta_j = 1, \theta_i \geq 0 \forall i) \quad (6)$$

$$\propto \prod_{j=1}^m \theta_j^{c_j + \alpha_j - 1} I(\sum_j \theta_j = 1, \theta_i \geq 0 \forall i) \quad (7)$$

This implies

$$\boldsymbol{\theta} \mid \mathbf{X} \sim \text{Dirichlet}(\mathbf{c} + \boldsymbol{\alpha}).$$

Takeaways

1. Dirichlet is conjugate for Categorical or Multinomial.⁴
2. Useful formula:

$$\prod_i \text{Multinomial}(x_i \mid \theta) \times \text{Dir}(\theta \mid \alpha) \propto \text{Dir}(\theta \mid \mathbf{c} + \alpha).$$

⁴The word Categorical seems to be used in CS and ML. The word Multinomial seems to be used in Statistics and Mathematics.